Chapter 3.2 — Formula Sheet

Market Microstructure & Order Book Fair Value

Companion quick-reference to your Notebook 3 syllabus and your *20 Worked Examples* set (Example labels 1–20 cross-referenced below)

Fair Value Fundamentals

Core Definition & Linearity [Ex 1–4, 6]

$$\text{Fair value} = \mathbb{E}[\text{payoff}] = \sum_i (\text{outcome}_i \times \mathbb{P}(\text{outcome}_i))$$

Linearity (no independence needed, chains across any number of pieces):

$$\mathbb{E}\left[\sum_i a_i X_i + b\right] = \sum_i a_i \mathbb{E}[X_i] + b$$

Independence shortcut (product of independent payoffs):

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] \quad (X \perp Y)$$

Law of Total Expectation [Ex 3]

Condition on a branch/regime B:

$$\mathbb{E}[X] = \sum_b \mathbb{P}(B = b)\,\mathbb{E}[X \mid B = b]$$

Watch for divergence (Ex 2, St. Petersburg): $\sum$(payoff × prob) can fail to converge — "no finite fair value" is sometimes the correct answer, not a number.

Inventory Risk & Fill Probability

Skew Direction & Toy Fill Models [Ex 12–13]

Skew rule: long inventory ⇒ shift *both* quotes down (eager to sell); short ⇒ shift *both* up (eager to buy back). **Toy fill-probability model:**

$$\mathbb{P}(\text{fill}) = a - b \cdot d \quad (\text{floored at } 0), \quad d = |\text{quote} - \text{FV}|$$

Closer to fair value (smaller d) ⇒ higher fill probability, lower edge per fill — no free lunch.

Expected P&L, Net of Holding Cost [Ex 14–15]

$$\mathbb{E}[\text{P\&L}]_{\text{gross}} = \sum_i \left(\text{edge}_i \times \mathbb{P}(\text{fill}_i)\right)$$

$$\mathbb{E}[\text{P\&L}]_{\text{net}} = \mathbb{E}[\text{P\&L}]_{\text{gross}} - (\text{cost per unit}) \times |\text{inventory}|$$

Compute holding cost *last*, as its own term — never fold it into either side's quoted edge.

Reservation Price [Ex 16]

$$r = \text{FV} - q\gamma\sigma^2$$

q = signed inventory (long > 0), γ = risk-aversion, σ^2 = payoff variance. Center quotes on r, not raw FV:

$$\text{bid} = r - k\sigma, \qquad \text{ask} = r + k\sigma$$

Skew size scales *linearly* in both inventory *and* variance — a volatile contract punishes the same position size far more (Ex 20).

Bid, Ask, Spread & Mid

Core Quote Mechanics [Ex 7–9]

$$\text{spread} = \text{ask} - \text{bid}, \qquad \text{mid} = \frac{\text{bid} + \text{ask}}{2}$$

House convention (half-spread scales with uncertainty σ):

$$\text{half-spread} = k\sigma \;\Rightarrow\; \text{bid} = \text{center} - k\sigma, \quad \text{ask} = \text{center} + k\sigma$$

Mid = true fair value only when the quote is symmetric around it — otherwise mid reveals your skew (Ex 7).

NBBO, Crossed Markets & Adverse Selection [Ex 10–11]

Best bid $= \max_i(\text{bid}_i)$, best ask $= \min_i(\text{ask}_i)$ across makers. **Crossed market** (risk-free arb): if best bid > best ask,

$$\text{arb profit/unit} = \text{best bid} - \text{best ask}$$

Adverse selection: a fill is conditioned on someone wanting to trade with you — realized edge on fills is systematically worse than quoted edge, especially against informed flow.

Capstone: The Full Pipeline

From Contract to Quote, Four Steps [Ex 17, 19]

$$\mathbb{E}[X] \;\to\; \mathbb{E}[X^2] \;\to\; \text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \;\to\; \sigma = \sqrt{\text{Var}(X)}$$

$$\to\; \text{half-spread} = k\sigma \;\to\; \text{bid} = \mathbb{E}[X] - k\sigma, \quad \text{ask} = \mathbb{E}[X] + k\sigma$$

Speed-trick reminder: look for a closed-form identity (e.g. $\sum k^3 = (\sum k)^2$) before brute-force summing — recognizing the pattern beats raw arithmetic speed (Ex 19).

Sequential / Multi-Round Quoting [Ex 18, 20]

Each round: (1) update fair value, (2) recompute $r = \text{FV} - q\gamma\sigma^2$ from *current* inventory, (3) quote $\pm k\sigma$ (or ± fixed width) around r, (4) after a fill, update q and repeat.

$$\text{short} \Rightarrow r = \text{FV} + |q| \times (\text{skew rate})$$

Full round P&L $= \mathbb{E}[\text{P\&L}]_{\text{gross}} - (\text{holding cost})$, exactly as in the Inventory section — the capstone is these two sections run back-to-back, every round.

One-tool-tree reminder: fair value is always $\sum$(outcome × probability), computed fast via linearity and identities; a quote is a spread placed around an estimate, sized to uncertainty; inventory and fill probability pull that center to a reservation price that protects you from adverse selection. Every "hard" market-making question is just these three ideas, chained in the order: value it, quote it, skew it.